

Quantum Category $\mathcal{O}$, I

1. Quantum groups & their forms
2. Quantum category $\mathcal{O}$

Quantum Groups and Their Forms

Notation: $\mathfrak{g}$ simple Lie algebra/ $\mathbb{C}$, simply laced (ADE) — to simplify the exposition.

$\mathfrak{h} \subset \mathfrak{g}$ Cartan

$\Lambda_r \subset \Lambda \subset \mathfrak{h}^*$ root & weight lattice

α_i ($i \in I$) simple roots, $(a_{ij})_{i,j \in I}$ — Cartan matrix.

v : indeterminate $\leadsto$ field $\mathbb{C}(v)$

$(\cdot, \cdot)$: invariant form on $\mathfrak{h}^*$ normalized by $(\alpha_i, \alpha_i) = 2$.

Generic Version

Definition 1 (Quantum group $U_v(\mathfrak{g})$). $U_v(\mathfrak{g})$ is $\mathbb{C}(v)$ -algebra w. generators E_i, F_i ($i \in I$), K_ν ($\nu \in \Lambda$) & relations:

- $K_\nu K_{\nu'} = K_{\nu+\nu'}, K_0 = 1$
- $K_\nu E_i K_\nu^{-1} = v^{(\nu, \alpha_i)} E_i, K_\nu F_i K_\nu^{-1} = v^{-(\nu, \alpha_i)} F_i$
- $[E_i, F_j] = \delta_{i,j} \frac{K_{\alpha_i} - K_{\alpha_i}^{-1}}{v - v^{-1}}$
- quantum Serre relations: $E_i E_j = E_j E_i$ if $a_{ij} = 0$
 $E_i E_j^2 + E_j^2 E_i = (v + v^{-1}) E_j E_i E_j$ if $a_{ij} = -1$
& same for F_i 's

Forms and Specializations

We want to consider $\mathbb{C}[v^{\pm 1}]$ -forms of $U_v(\mathfrak{g})$ i.e. $\mathbb{C}[v^{\pm 1}]$ -subalgebras $A \subset U_v(\mathfrak{g})$ w.
 $\mathbb{C}(v) \otimes_{\mathbb{C}[v^{\pm 1}]} A \xrightarrow{\sim} U_v(\mathfrak{g})$

To define the forms we need we introduce some notation.

$n \in \mathbb{Z}_{\geq 0} \leadsto [n]_v = v^{n-1} + v^{n-3} + \dots + v^{-(n-1)}$, a quantum integer

$$[n]_v! = \prod_{i=1}^n [i]_v$$

$E_i^{(n)} = E_i^n / [n]_v!$ (divided power), same for $F_i^{(n)}$.

Definition 2 (Forms of $U_v(\mathfrak{g})$).

- The De Concini-Kac form $U_v^{DK}(\mathfrak{g})$ is the $\mathbb{C}[v^{\pm 1}]$ -subalgebra generated by E_i 's, F_i 's & K_ν 's (the most obvious form).
- The Lusztig form U_v^l : by $E_i^{(n)}, F_i^{(n)}, K_\nu$ ($\forall i \in I, \nu \in \Lambda, n \in \mathbb{Z}_{\geq 0}$)
- The mixed form U_v^{mix} : $E_i^{(n)}, F_i, K_\nu$ ($\forall i \in I, \nu \in \Lambda, n \in \mathbb{Z}_{\geq 0}$)

The point of considering the forms is that one can specialize v to $q \in \mathbb{C}^\times$ getting algebras over $\mathbb{C}$, e.g. $U_q^{mix} = U_v^{mix} \otimes_{\mathbb{C}[v^{\pm 1}]} \mathbb{C}_q$ ($\mathbb{C}_q$ is 1-dim. $\mathbb{C}[v^{\pm 1}]$ -algebra w. $v \mapsto q$). When q is not a root of 1, $[n]_q \neq 0 \forall n \Rightarrow U_q^{DK} = U_q^{mix} = U_q^l$. But when q is primitive $\sqrt[d]{1}$ we have $[d]_q = 0$ (if d is even, $[d/2]_q = 0$) & the specializations behave differently from each other.

There are several reasons to be interested in the specializations to $q = \sqrt[d]{1}$ from a representation-theoretic perspective. Historically first was a connection to modular (= mod p) representation theory of $U(\mathfrak{g})$ & the corresponding algebraic group G .

The form whose representation theory resembles that of $G/\overline{\mathbb{F}}_p$ is U_q^l , it's also the best understood (since the 90's). The form whose representations resemble those of $U(\mathfrak{g}_{\overline{\mathbb{F}}_p})$ is U_q^{DK} , it's least understood. For these lectures we'll concentrate on the intermediate form, U_q^{mix} .

Quantum Category $\mathcal{O}$

Definition

We fix an odd positive integer d . Let ε be a primitive $\sqrt[d]{1}$. Note that U_v^{mix} & U_ε^{mix} are graded by Λ_0 , $U_\varepsilon^{mix} = \bigoplus_{\lambda \in \Lambda_0} U_{\varepsilon, \lambda}^{mix}$

$$\deg F_i = -\alpha_i, \quad \deg K_\nu = 0, \quad \deg E_i^{(l)} = l\alpha_i$$

so it makes sense to speak about the category of graded U_ε^{mix} -modules:

- the objects are U_ε^{mix} -modules M w. direct sum decomposition $M = \bigoplus_{\mu \in \Lambda_0} M_\mu$ w.
 $U_{\varepsilon, \lambda}^{mix} M_\mu \subset M_{\lambda+\mu} \forall \lambda, \mu \in \Lambda_0$
- morphisms are U_ε^{mix} -linear maps $\varphi : M \rightarrow N$ w.

$$\varphi(M_\mu) \subset N_\mu \quad \forall \mu.$$

Definition 3 (Quantum category $\mathcal{O}$). The quantum category $\mathcal{O}$, to be denoted by $\mathcal{O}_\varepsilon^{\text{mix}}$, is the full subcategory in the category of graded $U_\varepsilon^{\text{mix}}$ -modules consisting of all M s.t.

1. M is finitely generated over $U_\varepsilon^{\text{mix}}$,
2. $K_\nu|_{M_\mu} = \varepsilon^{(\mu, \nu)} \text{id}_{M_\mu}$, where $(\cdot, \cdot)$ is a W -invariant form on $\mathfrak{h}^*$ normalized so that $(\alpha, \alpha) = 2 \forall$ root α – recall that we assume that $\mathfrak{g}$ is simply laced for simplicity.
3. $\{\mu \mid M_\mu \neq 0\}$ is bounded from above: $\exists \mu_M \in \Lambda_0$ s.t.
 $M_\mu \neq 0 \Rightarrow \mu_M - \mu \in \text{Span}_{\mathbb{Z}_{\geq 0}}(\alpha_i \mid i = 1, \dots, r).$

Example 1 (Verma module).

$$\lambda \in \Lambda_0 \leadsto \Delta_\varepsilon^{\text{mix}}(\lambda) = U_\varepsilon^{\text{mix}} / U_\varepsilon^{\text{mix}} \left(K_\nu - \varepsilon^{(\nu, \lambda)}, E_i^{(l)} \mid \forall i, \nu, l \right)$$

w. class of 1 in $\text{deg} = \lambda$

Remarks

1. Note that the definition makes perfect sense when ε is not a root of 1. In this case (ii) can be made into the definition of M_μ : $\varepsilon^{(\nu, \mu_1)} = \varepsilon^{(\nu, \mu_2)} \forall \nu \Rightarrow \mu_1 = \mu_2$. So the category $\mathcal{O}_\varepsilon$ (we omit the superscript "mix" b/c there's no difference between the three specializations) is the complete analog of the classical BGG category $\mathcal{O}$ of representations of simple Lie algebras.
2. A counterpart of $\mathcal{O}_\varepsilon^{\text{mix}}$ for modular representations is as follows. Let $\mathbb{F} = \overline{\mathbb{F}}_p$, $B_\mathbb{F} \subset G_\mathbb{F}$ be a Borel subgroup. We can talk about $(\mathfrak{g}_\mathbb{F}, B_\mathbb{F})$ -modules: $U(\mathfrak{g}_\mathbb{F})$ -modules M equipped in addition w. a rational representation of $B_\mathbb{F}$ s.t. the action map $U(\mathfrak{g}_\mathbb{F}) \otimes M \rightarrow M$ is $B_\mathbb{F}$ -linear & the differential of the $B_\mathbb{F}$ -action coincides w. the action of $\mathfrak{b}_\mathbb{F} \subset \mathfrak{g}_\mathbb{F}$. The modular analog of $\mathcal{O}_\varepsilon^{\text{mix}}$ consists of all $(\mathfrak{g}_\mathbb{F}, B_\mathbb{F})$ -modules finitely generated over $U(\mathfrak{g}_\mathbb{F})$.
3. As usual, each Verma $\Delta_\varepsilon^{\text{mix}}(\lambda)$ has a unique irreducible quotient, to be denoted $L_\varepsilon^{\text{mix}}(\lambda)$ & $\lambda \mapsto L_\varepsilon^{\text{mix}}(\lambda) : \Lambda_0 \xrightarrow{\sim} \text{Irr}(\mathcal{O}_\varepsilon^{\text{mix}})$. One can show that $\dim L_\varepsilon^{\text{mix}}(\lambda) < \infty$. On the other hand, $\Delta_\varepsilon^{\text{mix}}(\lambda)$ is infinite dimensional (in fact its formal character is the same as of the usual Verma). It follows that objects in $\mathcal{O}_\varepsilon^{\text{mix}}$ may have infinite length. But they are Noetherian – in fact, each object in $\mathcal{O}_\varepsilon^{\text{mix}}$ admits a finite filtration by quotients of Verma modules.

Block Decomposition of $\mathcal{O}_\varepsilon^{\text{mix}}$

It turns out that the category $\mathcal{O}_\varepsilon^{\text{mix}}$ introduced last time splits into the direct sum of subcategories called blocks (a more precise name would be "infinitesimal blocks"). To define the blocks we'll need an affine Weyl group W^a and its suitable action on Λ_0 , the root lattice of $\mathfrak{g}$.

We set $W^a = W \ltimes \Lambda_0$. We write t_μ for $\mu \in \Lambda_0$ viewed as an element of W^a . Define an action of W^a on Λ_0 by

$$\begin{aligned} w \cdot \lambda &:= w(\lambda + \rho) - \rho \quad (\text{as usual } \rho = \frac{1}{2} \sum_{\alpha > 0} \alpha) \\ t_\mu \cdot \lambda &:= \lambda + d\mu \quad (\text{where } d \text{ is the order of } \varepsilon) \end{aligned}$$

Definition 4. Let Θ be a W^a -orbit in Λ_0 . Define the full subcategory $\mathcal{O}_\varepsilon^\Theta \subset \mathcal{O}_\varepsilon^{\text{mix}}$ as consisting of all objects M that admit a (finite) filtration by quotients of $\Delta_\varepsilon(\lambda)$, $\lambda \in \Theta$.

Proposition 1. $\mathcal{O}_\varepsilon^{\text{mix}} = \bigoplus_{\Theta} \mathcal{O}_\varepsilon^\Theta$.

Sketch of proof. Recall the form $U_v^{DK} \subset U_v^{\text{mix}} \subset U_v(\mathfrak{g})$.

Exercise 1. The center Z_v^{DK} of U_v^{DK} is central in U_v^{mix} & consists of $\deg 0$ (w.r.t. the Λ_0 -grading on U_v^{mix}) elements.

So Z_v^{DK} acts on any object in $\mathcal{O}_\varepsilon^{\text{mix}}$ by endomorphisms & functorially leading to the decomposition of this category into the direct sum indexed by homomorphisms $Z_v^{DK} \rightarrow \mathbb{C}$. The following was essentially established by De Concini & Kac.

Fact 1. $Z_v^{DK} \xrightarrow{\sim} \text{Span}_{\mathbb{Z}[v^{\pm 1}]}(K_{2\nu} \mid \nu \in \Lambda)^{(W, \cdot)}$ & under this identification, $\sum_{\omega \in W} K_{2\omega \cdot \nu}$ acts on $\Delta(\lambda)$ by $\sum_{\omega \in W} \varepsilon^{2(\omega \cdot \nu, \lambda)}$

Exercise 2. Deduce Proposition from Fact.

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